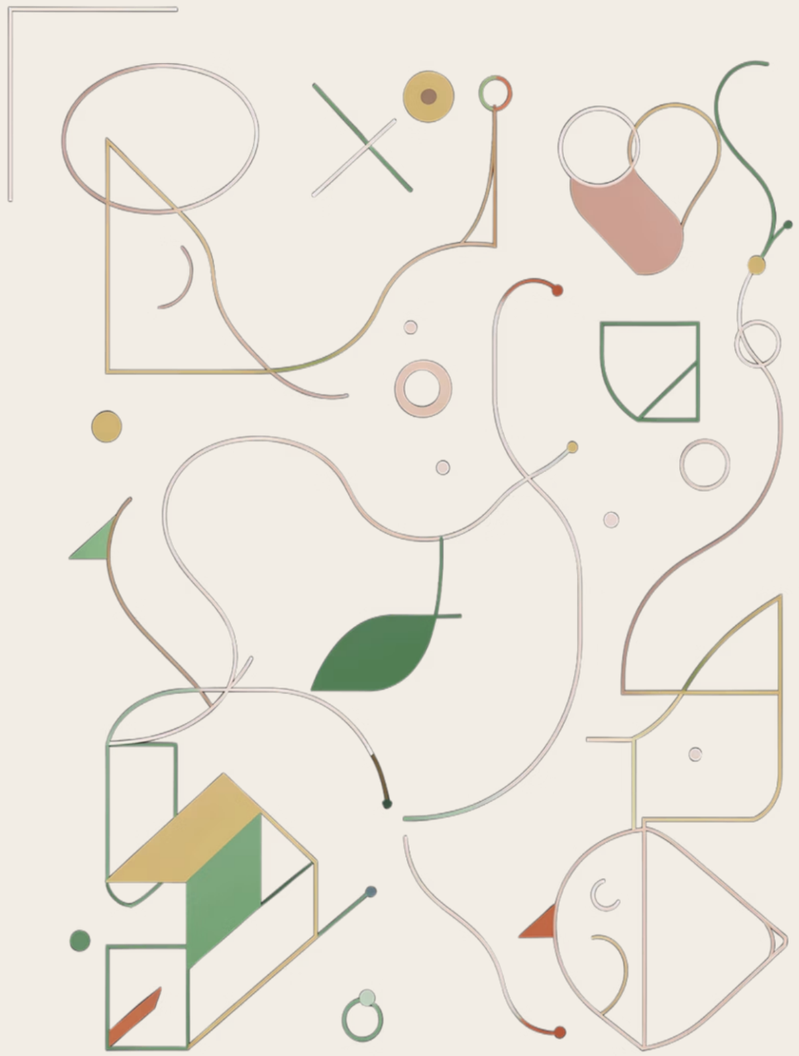


“(BE04000261)DISCERETE MATHEMATICS AND GRAPH THEORY”

PPT TOPIC: Combination and Pegionhole Priciple

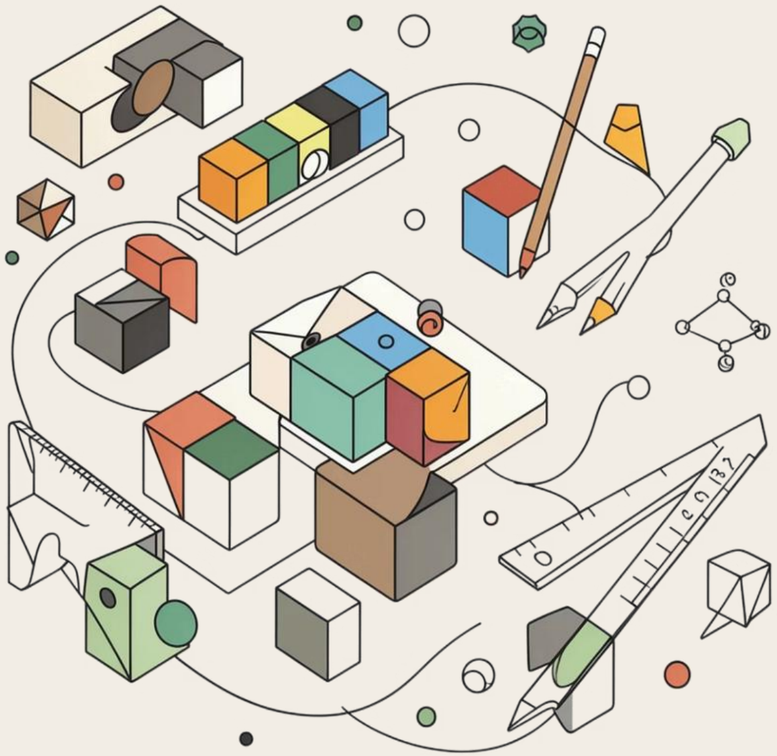
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Introduction to Combinatorics

Combinatorics is a branch of mathematics concerned with counting, arranging, and selecting objects. It provides systematic methods for answering questions like "How many ways can this happen?" — without having to list every possibility by hand.

Combination

Selecting items from a set **without regard to order**. Choosing Alice and Bob is the same as choosing Bob and Alice.

Pigeonhole Principle

A simple yet powerful counting principle stating that if you have more items than containers, at least one container must hold multiple items.

Why It Matters

These concepts are foundational in **computer science, cryptography, probability theory,** and algorithm design.

Topics Covered

This presentation walks through the core ideas of combinatorics step by step, from basic definitions to real-world applications.

#	Topic	Description
1	Introduction to Combinatorics	Overview of counting techniques and why they matter
2	Combination	Selection of items without considering order
3	Formula of Combination	Mathematical representation: nCr = n! / r!(n−r)!
4	Examples of Combination	Practical problems solved step by step
5	Pigeonhole Principle	Basic concept and intuitive examples
6	Generalized Pigeonhole Principle	Advanced form using the ceiling function
7	Applications	Real-life uses in CS, cryptography, and scheduling

CORE CONCEPT

What is a Combination?

A **combination** is a selection of items from a larger set where the **order does not matter**. This is the key distinction from permutations, where order is important.

Key Insight: Selecting students A and B is the *same combination* as selecting B and A. They count as one outcome, not two.

Combinations are used whenever the question asks: *"In how many ways can we choose…?"* rather than *"In how many ways can we arrange…?"*

Simple Example

Select **2 students** from a group of 5.

The pair {Alice, Bob} is identical to {Bob, Alice} — order is irrelevant.

This is fundamentally different from arranging them in a line, where AB ≠ BA.

Formula of Combination

The number of ways to choose **r** items from a set of **n** distinct items (without regard to order) is given by the binomial coefficient:

$$\binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$



n — Total Items

The size of the full set from which we are selecting. Must be a non-negative integer.



r — Items Selected

The number of items we choose. Must satisfy **0** ≤ **r** ≤ **n**.



! — Factorial

n! = **n** × **(n−1)** × **…** × **2** × **1**. By convention, **0!** = **1**.

Worked Example: Choosing 2 from 4 Students

Let's apply the formula to a concrete problem. Suppose we have 4 students: Alice, Bob, Carol, and David. How many ways can we choose a team of 2?

Step-by-Step Calculation

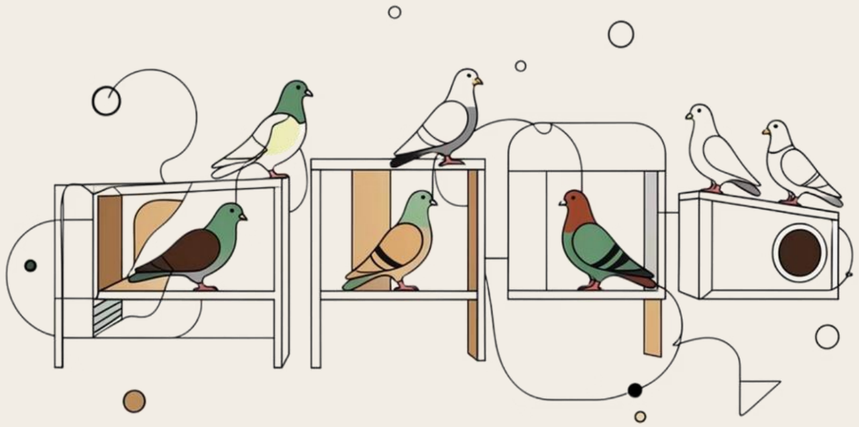
Here, **n** = **4** and **r** = **2**.

$$ {}_4C_2 = \frac{4!}{2!(4-2)!} = \frac{4!}{2! \times 2!} = \frac{24}{2 \times 2} = \frac{24}{4} = 6 $$

The 6 Combinations

1. Alice & Bob
2. Alice & Carol
3. Alice & David
4. Bob & Carol
5. Bob & David
6. Carol & David

Notice that {Alice, Bob} and {Bob, Alice} are counted only **once**.



The Pigeonhole Principle

One of the most elegant ideas in discrete mathematics — simple to state, surprisingly powerful in application.

> If **n** items are placed into **m** boxes and **n** > **m**, then at least one box must contain **more than one item**.

The Setup

You have more objects (pigeons) than containers (holes).

The Guarantee

No matter how you distribute them, at least one container holds ≥ 2 items.

The Power

This non-constructive proof technique guarantees existence without finding the exact arrangement.

Example: 10 Socks in 9 Drawers

The Problem

You have **10 socks** and only **9 drawers** to store them in.

No matter how carefully you distribute the socks, at least **one drawer must contain 2 or more socks**.

Here, **n** = **10** (items) and **m** = **9** (boxes). Since 10 > 9, the principle applies directly.

Why This Works

Imagine trying to avoid the conclusion — you place one sock in each of the 9 drawers. You still have **1 sock left**. Wherever you put it, that drawer now has 2 socks.

> This is a **non-constructive proof**: we know a drawer has 2 socks without knowing *which* drawer it is.

This same logic applies to many real problems: birthdays in a room, hash collisions, seating arrangements, and more.

Generalized Pigeonhole Principle

The basic principle tells us at least one box has 2 items. The **generalized** version tells us exactly how many items are guaranteed in the fullest box.

> If **n** objects are placed into **m** boxes, then at least one box contains **⌈n/m⌉ objects**, where ⌈ ⌉ is the ceiling function (round up to the nearest integer).

Compute n ÷ m

Divide the number of objects by the number of boxes to get the average load per box.

Apply Ceiling ⌈ ⌉

Always round *up*. Even 3.01 becomes 4. This gives the guaranteed minimum in the fullest box.

Interpret the Result

At least one box is guaranteed to hold that many items — no distribution can avoid it.

Example: 25 students in 4 groups → ⌈25/4⌉ = ⌈6.25⌉ = **7**. At least one group has 7 or more students.

Applications in the Real World

These principles are not just theoretical — they underpin critical systems in technology and mathematics.

Computer Science

Hash tables, data storage, and load balancing rely on pigeonhole reasoning to handle collisions and distribute data.

Cryptography

Combinations determine the size of key spaces. The pigeonhole principle proves that collisions are inevitable in hash functions.

Scheduling

Proving that conflicts are unavoidable when tasks exceed available time slots uses the generalized pigeonhole principle.

Error Detection

Checksums and parity bits use combinatorial counting to detect transmission errors in digital communication.

Mathematical Proofs

Both concepts are powerful tools in existence proofs — showing something *must* exist without constructing it explicitly.

Key Takeaways

Combinations: Order Doesn't Matter

Use **nCr = n! / r!(n−r)!** when selecting items and the arrangement is irrelevant.

Pigeonhole Principle: Guaranteed Overlap

If **n > m**, at least one box holds **⌈n/m⌉** items — a powerful non-constructive proof tool.

Both Are Widely Applicable

From cryptography to scheduling, these concepts are essential tools in discrete mathematics and computer science.

Thank You